Networks and Complexity

Solution 20-3

This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at https://github.com/NC-Book/NCB

Ex 20.3: 5-chain [Lvl. 2]
Find the eigenvalues of the adjacency matrix of a chain of five nodes.

Solution
We assume that the nodes are numbered from one end of the chain, i.e.

(1)-(2)-(3)-(4)-(5)

The orbit structure is (15)(24)(3), hence there are three eigenvectors with even symmetry and two with odd symmetry.

We start by finding the eigenvalues that belong to the eigenvectors with odd symmetry. There are two ways to find the eigenvalues. First, the more boring option: Since the eigenvector elements need to sum to zero in each orbit, the eigenvector has the structure

$$\boldsymbol{v} = (a, b, 0, -b, -a)^{\mathrm{T}}. \tag{1}$$

Multiplying $\mathbf{A}$

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ 0 \\ -b \\ -a \end{pmatrix} = \begin{pmatrix} b \\ a \\ 0 \\ -a \\ -b \end{pmatrix} \tag{2}$$

So, this is $\mathbf{A}\boldsymbol{v}$. If $\boldsymbol{v}$ is an eigenvector, then this should be the same as $\lambda\boldsymbol{v}$, which yields the condition

$$\begin{pmatrix} b \\ a \\ 0 \\ -a \\ -b \end{pmatrix} = \lambda \begin{pmatrix} a \\ b \\ 0 \\ -b \\ -a \end{pmatrix} \tag{3}$$

Due to the symmetry the condition in the third row is trivial and row 5 and 4 contain the same condition as 1 and 2. Hence the equation stipulates only two independent conditions

$$b = \lambda a \tag{4}$$
$$a = \lambda b \tag{5}$$

If we substitute one of these equations into the other we find

$$a = \lambda^2 a \tag{6}$$

which implies

$$\lambda^2 = 1 \tag{7}$$

and hence

$$\lambda = \pm 1 \tag{8}$$

Could we have found this result more quickly? Sure. Just delete the central node from the chain. This leaves us with a network consisting of two linked pairs:

(1)-(2) (4)-(5)

Maybe you still remember that the linked pair has eigenvalues 1 and -1. Otherwise, these are very quick to find. Because we have two pairs, the eigenvalues of the reduced network are

$$\mu_1 = 1, \quad \mu_2 = 1, \quad \mu_3 = -1, \quad \mu_4 = -1. \tag{9}$$

By the interlacing theorem, the eigenvalues of the original network have to interlace with these, which also reveals that 1 and -1 must have been eigenvalues of the original network.

Now for the eigenvectors with even symmetry. These eigenvectors must have the form $(a, b, c, b, a)^{\mathrm{T}}$, and hence applying the eigenvector equation gives us

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ b \\ a \end{pmatrix} = \begin{pmatrix} b \\ a+c \\ 2b \\ a+c \\ b \end{pmatrix} = \lambda \begin{pmatrix} a \\ b \\ c \\ b \\ a \end{pmatrix} \tag{10}$$

This gives us three conditions

$$b = \lambda a \tag{11}$$

$$a + c = \lambda b \tag{12}$$

$$2b = \lambda c \tag{13}$$

One solution is $\lambda = 0$ which implies $b = 0$ and $a = -c$. (Not the interesting symmetry of this eigenvector, we'll come back to it later) Furthermore, comparing the first and the third equation, we see that

$$c = 2a \tag{14}$$

Substituting into the second equation and then using the first again gives us

$$3a = \lambda b = \lambda^2 a \tag{15}$$

Therefore,

$$\lambda^2 = 3, \tag{16}$$

so the last two eigenvalues are

$$\lambda = \pm\sqrt{3}. \tag{17}$$

Could we have found the three even-symmetry eigenvalues more quickly? Sure. From the odd-symmetry eigenvectors we already know the eigenvalues 1 and -1 The network is bipartite, which means that the three missing eigenvalues must be 0 and a symmetric pair. Let's say the symmetric eigenvalues are m and $-m$. We know that the squares of all eigenvalues must add up to two times the number of links, so

$$(1)^2 + (-1)^2 + 0^2 + (m)^2 + (-m)^2 = 2 \cdot 4 \tag{18}$$

and hence

$$2m^2 = 6 \tag{19}$$

so $m = \sqrt{3}$